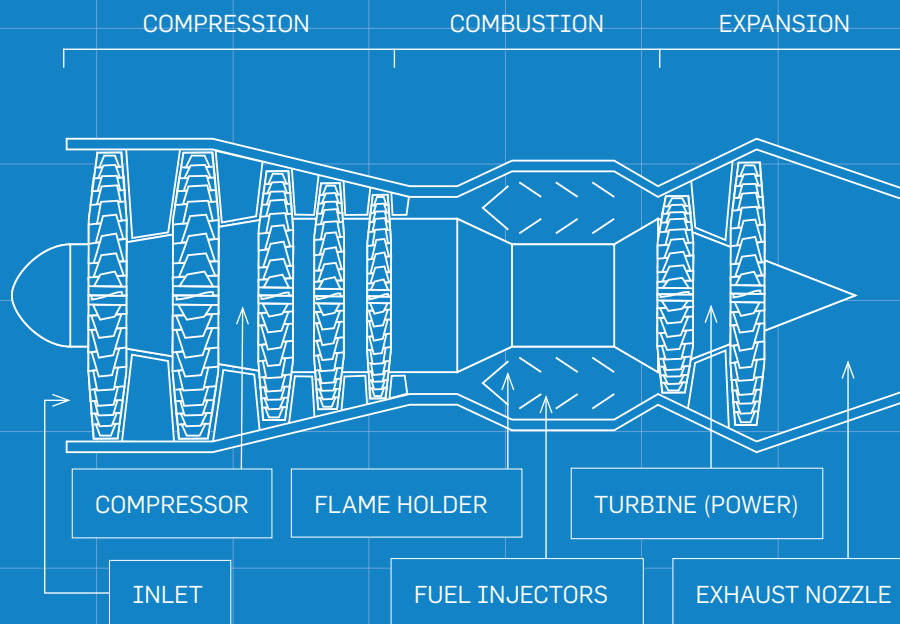


TYPE	TURBOJET
SPEED	MACH 0 - 3



The essential principle behind the functioning of this jet engine is the same as that of the RAMJET. The difference is in the design, where turbines have to do the job of compression. A compressor in the front of the engine sucks in air and compresses it. The air then enters the combustion zone. The hot air ignites the injected fuel and is ejected out the back when it expands. Before being ejected however, the air is passed over another turbine which uses the kinetic energy of the air to generate electricity as well as drive the compressor shaft, increasing efficiency.

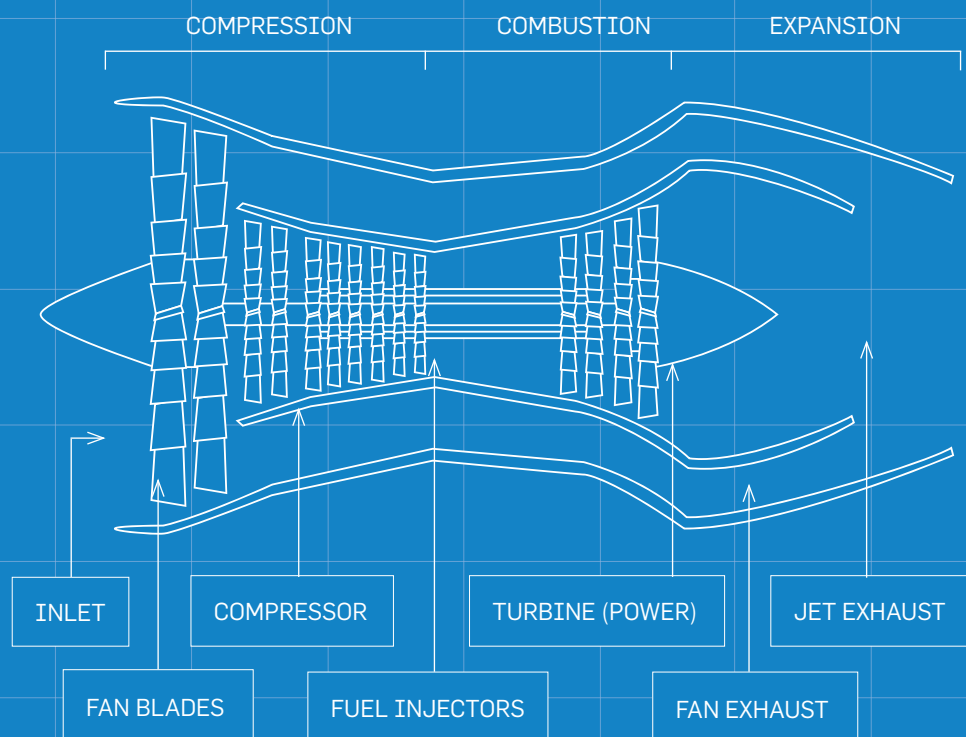
The advantage of this engine is that it can take an aircraft from a stand-still to a very high velocity. The only drawback is that the engine, though capable, is inefficient at low subsonic, and high supersonic speeds. The sweet spot for efficiency is around Mach 1 - 1.5 and is thus, very suitable for current military fighter aircraft.

TYPE	TURBOFAN
SPEED	MACH 0 - 1.5

This is the most common jet engine in the world and is present on almost every modern aircraft currently in existence. You can call it a cross between a propeller driven aircraft and a jet. In fact, a variant of this engine (called a turboshaft engine) is used in helicopters and even the US military's famous M1A1 Abrams tank.

It works the same way as a TURBOJET except for one key difference- there's a fan present in front of the compressor. The function of this fan is to efficiently provide thrust at low speeds. You can think of it as a propeller added to a TURBOJET to improve its efficiency at subsonic speeds. The fan sucks air into the compressor as well as into a ducted passage designed to accelerate the air. The air that goes into the compressor undergoes the same process of compression, combustion and expansion as the TURBOJET.

The advantage of this design is that at very low speeds, when the incoming air pressure is insufficient for jet propulsion, the fan functions as a propeller. At higher speeds, it augments the TURBOJET, improving its efficiency at that speed and thus, the engine's overall efficiency. The disadvantage of this design is that an aircraft won't be capable of efficient supersonic flight owing to the drag induced by the fan at such speeds. These specifications are sufficient for commercial flights though.



NOTES	All figures not to scale
	*Mach: The speed of an object relative to the speed of sound
	Jet engines are also known as "reaction engines" as the reaction to the flow of exiting gases provides the required thrust
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